

Scalability & Future Plan

Date	30 JUNE 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority
1	Static dataset	Predictions limited to training data	High
2	Local deployment	Limited accessibility	Medium
3	No live weather integration	Less accurate recommendations	High

Scalability Plan:

S.No	Aspect	Current State	Proposed Upgrade
1	User Load	Single server	Deploy on cloud using AWS/Azure
2	Data Storage	SQLite/MySQL	Cloud Database
3	Performance	Single Flask instance	Load balancing & caching
4	Security	Basic authentication	JWT authentication & HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Weather API Integration	Weather API Integration	Weather API Integration
Phase 2	Fertilizer Recommendation	Fertilizer Recommendation	Fertilizer Recommendation
Phase 3	Mobile Application	Mobile Application	Mobile Application
Phase 4	IoT Sensor Integration	IoT Sensor Integration	IoT Sensor Integration